Databases Continuous Assessment 2
ER modelling - Ryanair

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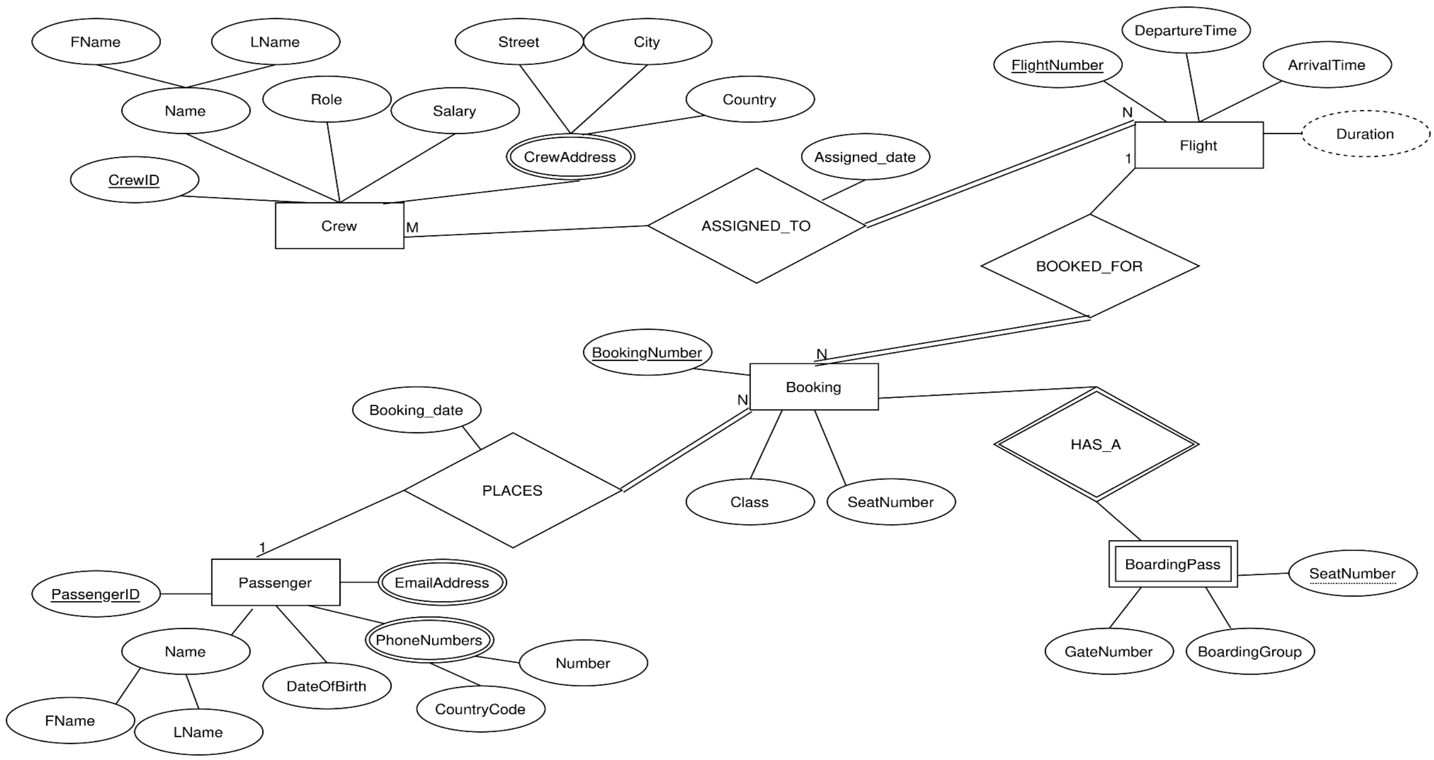
Task 1



Justification of selected attributes:

1. **Crew**
   a. CrewID: It is an atomic key attribute unique for each crew entity
   b. Name: It is a composite attribute since it can be divided into Fname and Lname.
   c. Role: It is an atomic single-valued attribute that stores one indivisible role such as captain or first officer.
   d. Salary: It is an atomic single-valued numeric attribute.
   e. CrewAddress: It is a **complex attribute** since a crew member can have multiple addresses, and each address can be divided into Street, City, and Country.
2. **Flight**
   a. FlightNumber: It is an atomic key attribute unique for each flight entity.
   b. DepartureTime: It is an atomic single-valued attribute.
   c. ArrivalTime: It is an atomic single-valued attribute.
   d. Duration: It is a **derived** attribute calculated from ArrivalTime and DepartureTime.
3. **Passenger**
   a. PassengerID: It is an atomic key attribute unique for each passenger entity.
   b. Name: It is a composite attribute since it can be divided into Fname and Lname.
   c. DateOfBirth: It is an atomic single-valued attribute.
   d. EmailAddress: It is a **multivalued** attribute since one passenger can register multiple emails
   e. PhoneNumbers: It is a **complex attribute** since a passenger may have more than one phone number, and each phone number can be divided into CountryCode and number.
4. **Booking**
   a. BookingNumber: It is an atomic key attribute unique for booking entity.

b. SeatNumber: It is an atomic single-valued attribute.
c. Class: It is an atomic single-valued attribute such as economy, priority, etc.

5. **BoardingPass (weak entity)**
   - **a.** SeatNumber: It is a partial key that uniquely identifies a BoardingPass within the same Booking entity.
   - **b.** BoardingGroup: It is an atomic single-valued attribute.
   - **c.** GateNumber: It is an atomic single-valued attribute.

Task 2

Weak entity

BoardingPass is modelled as weak because it does not contain a unique key attribute of its own. The same seat number might exist in multiple different bookings, thus SeatNumber attribute is not enough to uniquely identify BoardingPass entities in its entity set. Consequently, BoardingPass is a weak entity that depends on a strong owner entity to be identified. The strong entity that helps distinguish BoardingPass entities is Booking since every boarding pass is associated with a booking and it cannot exist independently without it. The owner entity (Booking) helps identify BoardingPass by using a combination of its key attribute and the partial key of BoardingPass.

The partial key of BoardingPass is SeatNumber which uniquely identifies a BoardingPass within the same Booking entity. The entire boarding pass identification for BoardingPass is then {BookingNumber, SeatNumber}.

Participation constraints

In the identifying relationship “Has_a”, with cardinality ratio 1:1, that connects Booking and BoardingPass, the corresponding participation constraints are:

- Booking: <u>partial participation</u> since there can be bookings without a boarding pass i.e. a passenger never does their check-in to obtain a boarding pass
- BoardingPass: <u>total participation</u> since a boarding pass cannot exist without a corresponding booking

Task 3

- Booking → “Has_a” → BoardingPass is **1:1**
  Cardinality ratio is 1:1 because each booking has one boarding pass and a boarding pass belongs to one booking.
  Participation constraints are as defined above.
- Passenger → “Places” → Booking is **1:N**
  Cardinality ratio is 1:N because one passenger can have multiple bookings, but each booking only belongs to one passenger.
  Participation constraints:
  - Passenger: <u>partial participation</u> because a passenger might not have any bookings yet
  - Booking: <u>total participation</u> since every booking requires its corresponding owner/passenger, without someone that booked a flight, booking doesn’t exist

  Relationship attributes:
  - Booking_date: It is an atomic single-valued attribute.
- Crew → “Assigned_to” → Flight is **N:M**
  Cardinality ratio is N:M because one crew member can be assigned to work on multiple flights and a flight can have multiple crew members.
  Participation constraints:
  - Crew: <u>partial participation</u> since some crew entities might be off
  - Flight: <u>total participation</u> because every flight entity needs at least one crew member

Relationship attributes:

- Assigned_date: It is an atomic single-valued attribute.

- Booking → “Booked_for” → Flight is **N:1**
  Cardinality ratio is N:1 because a flight can have multiple bookings, but all those bookings are just for one specific flight
  Participation constraints:
  - Booking: total participation because every booking must refer to a flight
  - Flight: partial participation since a flight can exist without a booking

Task 4

Step 1 – Mapping Regular Entity Types:

Composite attributes were replaced by their simple components.

Crew

CrewID	Fname	Lname	Role	Salary

Flight

FlightNumber	DepartureTime	ArrivalTime

Passenger

PassengerID	Fname	Lname	DateOfBirth

Booking

BookingNumber	SeatNumber	Class

Step 2 – Mapping Weak Entity Types and Step 3 - Mapping of Binary 1:1 Relationship Types:

Including PK of Booking (owner entity type) as FK of BoardingPass, renamed BookingNumber to BRNumber. BoardingPass PK is {BRNumber, SeatNumber}. This also corresponds to the mapping of the identifying 1:1 relationship.

Crew

CrewID	Fname	Lname	Role	Salary

Flight

FlightNumber	DepartureTime	ArrivalTime

Passenger

PassengerID	Fname	Lname	DateOfBirth

Booking

BookingNumber	SeatNumber	Class

BoardingPass

BRNumber	SeatNumber	BoardingGroup	GateNumber

Step 4 - Mapping of Binary 1:N Relationship Types:

Mapping relationships “Places” and “Booked_for” where entity Booking is the N-side entity in both. PassengerID attribute from Passenger entity is renamed to Pid in Booking entity. FlightNumber and Pid are now FKs in Booking

Crew

CrewID	Fname	Lname	Role	Salary

Flight

FlightNumber	DepartureTime	ArrivalTime

Passenger

PassengerID	Fname	Lname	DateOfBirth

Booking

BookingNumber	SeatNumber	Class	FlightNumber	Pid	Booking_date

Step 5 - Mapping of Binary M:N Relationship Types:

Created a new table mapping the relationship “Assigned_to” with relationship attribute Assigned_date and the PKs of Crew and Flight as FKs. The primary key of Assigned_to is {CrewID, FlightNumber}

Crew

CrewID	Fname	Lname	Role	Salary

Flight

FlightNumber	DepartureTime	ArrivalTime

Passenger

PassengerID	Fname	Lname	DateOfBirth

Booking

BookingNumber	SeatNumber	Class	FlightNumber	Pid	Booking_date

BoardingPass

BRNumber	SeatNumber	BoardingGroup	GateNumber

Assigned_to

CrewID	FlightNumber	Assigned_date

Step 6 - Mapping of Multivalued Attributes:

Created tables Passenger_Email, Passenger_PhoneNumbers, CrewAddress. Note: derived attributes (age, duration) are not stored as columns because they can be computed

Crew

CrewID	Fname	Lname	Role	Salary

Flight

FlightNumber	DepartureTime	ArrivalTime

Passenger

PassengerID	Fname	Lname	DateOfBirth

Booking

BookingNumber	SeatNumber	Class	FlightNumber	Pid	Booking_date

BoardingPass

BRNumber	SeatNumber	BoardingGroup	GateNumber

Assigned_to

CrewID	FlightNumber	Assigned_date

Passenger_Email

PassengerID	EmailAddress

Passenger_PhoneNumbers

PassengerID	CountryCode	Number

CrewAddresses

CrewID	Street	City	Country